MIL-R-38162B (USAF)
11 November 1971
SUPERSEDING
MIL-R-38162A (USAF)
11 January 1971

MILITARY SPECIFICATION
RADIOSONDE SET AN/AMT-13 ( )

1. SCOPE

*1.1 Scope. This specification covers one type of equipment, designated Radiosonde Set AN/AMT-13( ), for obtaining and transmitting upper air meteorological data to airborne receiving and recording equipment. The descending radiosonde collects the weather data between the flight level and the ground when ejected from the aircraft system. This radiosonde is designed to be used in conjunction with the AN/AMQ-25 and AN/AMQ-29 meteorological systems.

*2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture and Fungus Resistant, of Communications, Electronic and Associated Electrical Equipment
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specification and Standard, Order of Precedence for the Selection of
MIL-STD-415	Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. This specification makes provision for first article testing.

*3.2 Components. The radiosonde shall consist of:

ITEM NO.	QUANTITY	DESCRIPTION	SEE REQUIREMENT
1	1	Antenna	3.7.4.3
2	1	Sensor Mechanism	3.7.1
3	1	Commutator	3.7.2
4	1	Battery Housing	3.7.3
5	1	Transmitter	3.7.4
6	1	Accessory Kit	3.7.5

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials, parts, and components used in the manufacture of the radiosonde shall be of the best quality consistent with the service for which they are intended.

3.4.1 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion caused by atmospheric conditions likely to be met in storage or normal service use.

3.4.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.3 Fungus-proof materials. Fungus-proof materials shall be used where possible. Those materials used which are fungus nutrients shall be treated with fungustatic material in accordance with MIL-T-152, Type II.

3.4.4 Protective treatement. When materials are used in the construction of the radiosonde that are subject to deterioration when exposed to environmental conditions likely to occur during service life, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Any protective coating that will crack, chip or scale with age or extremes of climatic and environmental conditions shall not be used.

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3.5 Design and construction. Design of the radiosonde shall include provisions which prevent reversed installation of connectors, fasteners, parts, and units. Particular attention shall also be given to weight and interchangeability. All parts of the equipment shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, and operation without deterioration, malfunction, or any other damage that will affect its use.

3.5.1 Compatibility. The radiosonde shall be designed for successful actuation and launch from aircraft utilizing the AN/AMQ-25 and AM/AMQ-29 meteorological systems. It shall be compatible with Baseline Calibration Set AN/AMM-6 and Battery Charger-Radiosonde Case Group AN/AMA-3.

3.5.2 Maintenance. The radiosonde shall have a minimum number of parts consistent with item reliability and shall be contained in a single assembly or a minimum number of assemblies. The design shall utilize readily removable and replaceable units to facilitate easy assembly. The design shall be such that adjustments, minor repairs, and replacement of parts may be made by service personnel with a minimum of training and using common commercial tools.

3.5.3 Test points and built-in test facilities. Test points and built-in test facilities shall be included in the design of the equipment, if needed. These shall be simple in design, shall not cause protrusions in the housing of the radiosonde, and shall require a minimum of accessory equipment or special tools. Electronic test points shall be in accordance with MIL-STD-415.

*3.5.4 Storage life. The radiosonde shall be designed for minimum storage life of four years.

*3.6 Performance. The radiosonde shall be capable of meeting the requirements specified herein under the following conditions:

a. Ambient temperature from -80°C to +50°C.

b. After prolonged storage at any of all temperatures within the range of -65°C to +72°C.

c. Relative humidity up to 100 percent in combination with temperatures of -40°C to +50°C.

d. Pressure altitudes from 150 millibars (mb) to 1060 mb.

e. Vibration over the range of 10Hz to 55 Hz at a total double amplitude (DA) of 0.04 inch.

f. Acceleration of 5g maximum along centerline of cylinder only.

g. One shock of 75g maximum along centerline of cylinder.

h. Rain at 2 inches per hour and wind velocities up to 100 feet per second.

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i. Frost formation on and within the radiosonde.

3.6.1 Sand and dust. As a design objective the equipment, in both operating and non-operating conditions, shall withstand exposure to sand and dust particles as encountered in desert areas. No test shall be required.

3.6.2 Simulated flight conditions. The radiosonde shall demonstrate no abnormalities when subjected to simulated flight conditions. These shall include intermittent RF power output, shift to RF or audio frequencies outside tolerance limits, pressure sensor reversals, or change of comutation rate outside tolerance limits.

3.6.3 Measurement capabilities. The radiosonde shall measure, in the operational environment, the following:

a. Pressure from 1060 to 150 mb.
b. Temperature from -80°C to +50°C
c. Relative humidity from 10 to 100 percent in combination with temperatures of -40°C to +50°C.

3.6.4 Rate of descent. The radiosonde shall descend from an altitude of 40,000 feet to the surface in 8 minutes ± 1 minute.

*3.6.5 Audio curve. The frequency versus resistance relationship of each radiosonde (after reference correction and lock-in) shall conform to the characteristics of Table I. The tolerance shall be such that the temperature or pressure errors caused by any deviation from the standard curve are less than 0.25°C and 0.5 millibar (mb) respectively.

TABLE I
AUDIO FREQUENCY CURVE

Frequency	Resistance
6050	0 ohms
5483	10,000 ohms
4507	40,000 ohms
3886	80,000 ohms
3199	200,000 ohms
3149	218,000 ohms
2785	500,000 ohms

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3.7 Details of components.

3.7.1 Sensor mechanisms. The sensor mechanisms shall conform to the following:

*3.7.1.1 Pressure sensor. The pressure unit shall operate satisfactorily over the range of 1060 mb to 150 mb. The pressure sensor shall produce a minimum audio frequency of 2640 Hz at 1060 mb. The minimum resolution shall be 4.0 Hz per mb from 1060 mb to 950 mb and 2.0 Hz per mb from 950 mb to 150 mb. If a pressure capsule is used, it shall be aged for a minimum of 30 days prior to calibration. The date of manufacture shall be stamped on the capsule with indelible ink and the calibration date shall be stamped on the calibration data.

*3.7.1.2 Temperature sensor. The temperature sensor shall operate satisfactorily over the range of -80°C to +50°C. It shall not be adversely affected by direct water contact, either liquid or frozen. The resistance value at 30°C shall be stamped on each individual sensor board package (3.7.5b) and the element shall not require additional baselining. The baseline value shall be accurate to within 0.25°C at 30°C.

*3.7.1.3 Humidity sensor. The humidity sensor shall operate satisfactorily over the range of 10 to 100 percent relative humidity (RH) at all temperatures between -40°C and +50°C. If it is adversely affected by direct water contact, either liquid or frozen, it shall be protected by a shield. The resistance value at 33% RH and +25°C shall be stamped on each individual container and the element shall not require additional baselining. The baseline value shall be accurate to within ± 5% RH at the above conditions.

*3.7.2 Commutator. The commutator shall have the following circuits:

a. Twelve sampling periods shall represent: (1) high reference, (2) temperature, (3) humidity, (4) pressure, (5) temperature, (6) humidity, (7) low reference, (8) temperature, (9) humidity, (10), pressure, (11) temperature, (12) humidity.

b. Each of the twelve sampling periods shall be 300 milliseconds in length. A frequency of 2400 ± 50Hz shall be present between each sampling period for a duration of 50 milliseconds. The period times shall not vary within a given sonde to sonde by more than 10 percent.

*3.7.3 Battery housing. The battery housing shall allow for field replacement of the battery specified by the procuring activity. The design will allow replacement using ordinary tools and shall be compatible with existing connectors on the specified battery.

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*3.7.3.1 Battery switch. A single-pole single-throw switch shall control the power to the circuits of the radiosonde. This switch shall operate with an axial inward force of 5 1/2 pounds and stroke not exceeding 1/4 inch measured from the outside of the transmitter housing. The switch shall be compatible with the MX-8982/AMQ-25A and MX-9133/AMQ-31 radiosonde dispensers. In no case shall the switch actuator extend beyond the outer plane of the transmitter housing. This switch shall be a recycling type, capable of switching 1.5 amperes at the rated battery voltage.

*3.7.4 Transmitter characteristics. The transmitter shall consist of an audio frequency (AF) oscillator and a radio frequency (RF) oscillator. The AF oscillator shall pulse modulate the RF oscillator with a frequency proportional to the atmospheric parameter in accordance with paragraph 3.6.5. The RF oscillator shall provide RF energy to the antenna.

*3.7.4.1 Transmitter. The transmitter shall conform to the following:

a. RF oscillator frequency (post-launch). The frequency of the transmitter in the post-launch configuration shall be 403.0 MHz ± 1.5 MHz. The RF oscillator may be adjusted to this frequency.

b. RF oscillator frequency (pre-launch). The frequency of the transmitter in the pre-launch configuration shall be 403.0 MHz ± 2.5 MHz.

c. Power output. The transmitter shall deliver a minimum peak pulse power of 18 watts to the antenna at a pulse duration of 16 ± 3 microseconds.

*3.7.4.2 Audio frequency (AF) oscillator. The transmitter shall be modulated by an AF oscillator which develops pulses of 16 ± 3 microseconds duration. The pulse frequency of the AF oscillator shall range from 2350 Hz to 6110 Hz.

*3.7.4.2.1 Reference frequencies. The audio frequency (AF) oscillator shall provide a high and a low reference frequency. The high reference frequency shall be 6050 ± 60Hz and the low reference frequency shall be 3149 ± 50Hz.

*3.7.4.2.2 Audio frequency oscillator stability. The audio frequency oscillator shall have a short term stability of ± 5 Hz per minute, while operating.

3.7.4.3 Antenna radiation pattern. The E-plane pattern of the radiosonde antenna shall be two-lobed, with maximum intensity vectors directed upward at an angle of approximately 5 degrees (shall not exceed 20 degrees from the horizontal plane). The H-plane pattern shall approximate a circle around the radiosonde. The pattern shall be smooth and continuous, free from sharp gain variations, with a maximum variation of 3 db.

*3.7.5 Accessory kit. The accessory kit shall contain:

a. Humidity element, in accordance with 3.7.1.3.

b. Sensor holder, complete with temperature sensor in accordance with 3.7.1.2.

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3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.9 Instrument dimensions. The radiosonde (fully assembled with parachute container mounted and sensor head installed) shall be 18 ± 1/16 inches in length and 3 1/2 ± 1/64 inches in diameter measured at the widest point.

*3.10 Weight. The weight of the radiosonde, with batteries but less accessory kit, shall be less than 5 pounds.

*3.11 Nameplate. Nameplate data or identification information shall be permanently stenciled to the equipment. Stenciling shall be in accordance with MIL-STD-130.

*3.11.1 Serial numbers. Each radiosonde shall be serialized. The first four digits shall be the Julian date that the radiosonde is manufactured and the last five shall be a serialized number beginning with 00001. For example, a sonde manufactured on 15 September 1971 which happened to be the 125th unit completed on the production run would be 125800125.

*3.12 Government-loaned property. When the contract or purchase order so provides, and upon request by the contractor, the Government shall loan the following equipment contingent upon availability:

ITEM NO.	ITEM DESCRIPTION	QUANTITY
1	Radiosonde set AN/AMT-13 ( )	1
2	Radiosonde Case CY-4197/AMA-3	1
3	Charger, Battery PP-3924/AMA-3	1
4	Baseline Calibration Set TS-202/AMM-6	1
5	Receiver, Radio R-1196/AMQ-19	1
6	Antenna AT-896/A	1
7	Control Indicator C-3804/AMQ-25A	1

(NOTE: A thirty sonde battery charger may be supplied in lieu of items 2 and 3.)

3.13 Pressure sensor calibration data. Calibration data for the pressure sensor of each radiosonde shall be packed with the radiosonde. One copy of the data shall be typewritten on a sheet of paper. Two copies shall be provided in punched tape form. The tape shall be standard 7/8 inch wide, buff color oiled paper teletype tape. The data shall be punched in standard 5-level Baudot code with edge printing of the data. The data format shall be as follows:

a. Minimum 8 inches of space
b. Radiosonde serial number
c. Minimum of 4 inches of space (drive holes only)

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d. Pressure sensor resistance value in ohms. Values will be given for 1060mb, 1030mb, 1000mb, 950mb, and each 50mb up through 150mb. Each value shall use six digits with no spaces or punctuation between values. Numbers shall be right adjusted in the six digit field using zeroes to fill in on the left. Example: 23,900 ohms shall be punched as 023900. Values for 1060, 1030, 1000,...150mb shall follow in succession in the format of 0093800161000-23900...

e. Minimum of 8 inches of space (drive holes only)

f. The date of calibration of the pressure sensor shall be stamped or edge printed on the tape in the 4 inches of space between the serial number and the pressure data. The Julian date shall be used, i.e., 25 July 1969 would be 9206.

3.14 Workmanship. The radiosonde, including all parts and accessories, shall be fabricated and finished in a workmanlike manner with particular attention to burrs, sharp edges, accuracy of dimensions, thoroughness of soldering, welding, painting and wiring, alignment of parts, and tightness of assembly screws.

*3.15 Battery. The battery to be used shall be specified by the procuring activity. Batteries required for testing the radiosondes shall be government furnished.

*3.16 Standard, commercial and qualified parts. MIL-Standard and Army/Navy standard parts shall be used when they suit the purpose intended.

*3.17 Inspection of preparation for delivery. Inspection of preservation, packaging, and packing shall be accomplished in accordance with the quality assurance provisions of MIL-STD-794 to insure conformance to the requirements of section 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (See 4.4)
b. Quality conformance inspection (See 4.5)

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4.3 Test conditions. Test conditions shall be as specified in the applicable test.

*4.3.1 Pretest performance record. Prior to conducting any of the tests, the test item shall be operated at ambient conditions. A record shall be made of all data to insure compliance with required performance (including calibration of the pressure sensor, adjustment of RF and audio frequencies, etc.). This data shall provide the criteria for checking satisfactory performance during and at the conclusion of the tests.

*4.3.2 Failure criteria. Operation outside any of the following performance tolerance limits shall constitute a failure of the test item. All tests for these limits shall be performed with the antenna in the post-launch configuration.

a. Audio frequency oscillator (AF)high reference frequency short term stability: ± 5 Hz per minute.

b. AF oscillator high reference shift: ± 60 Hz

c. RF frequency short term stability: ± 1 MHz per minute (after a two-minute warmup).

d. RF frequency tolerance: ± 1.5 MHz at 403.0 MHz.

e. Audio curve shift limits specified in Table I after reference correction in accordance with the following formula:

(NOTE: Sensor resistances shall be replaced by precision decade resistances having ± 0.02 percent accuracy while making reference corrections)

$$\text{reference correction} = \frac{(F_x - F_2)(\Delta F_1 - \Delta F_2)}{F_1 - F_2} + \Delta F_2$$

WHERE
- F_1 = frequency at standard high reference
- F_2 = frequency at standard low reference
- ΔF_1 = difference between actual high reference and standard high reference
- ΔF_2 = difference between actual low reference and standard low reference
- F_x = any frequency on the new curve.

f. RF peak power output: not less than 18 watts at both high and low reference when measured on peak reading wattmeter or computed from the following formula when measured on an average power reading wattmeter:

$$P_p \text{ (in watts)} = \frac{r\,P_b}{ft} \text{ where}$$

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r = actual power ratio of R.F. attenuator used
P_b = average power measured in watts
f = R O frequency (pulse repetition rate) in Hz
t = R F pulse width in microseconds

g. R F pulse width: The transmitted pulses shall have a pulse width of 16 ± 3 microseconds when measured at the high reference frequency under the following conditions: A receiving antenna shall be loosely coupled to the radiosonde antenna. The received radiosonde signal shall be detected and displayed on an oscilloscope, using internal synchronization. The oscilloscope sweep shall be measured at 70 percent of peak value.

*4.3.3 Pressure sensor accuracy/repeatability. The procedure used to obtain the pressure sensor calibration data shall be repeated with the sensor installed in the radiosonde. The data obtained shall be within ± 3 millibars of the original calibration data at each point.

*4.3.4 Current drain. The total radiosonde circuit current drain shall meet the following conditions:

a. The current measured at the battery terminals shall be less than 0.6 ampere.
b. Battery voltage not less than 15 percent below rated voltage after operating the sonde for 25 minutes.

4.3.5 Operating power. The radiosonde batteries shall be recharged (or replaced, as required) before performing any and all tests which require operation from internal power. External power sources used to operate the radiosonde shall be regulated to within one percent of the supply voltage.

4.4 First article inspection.

*4.4.1 Test samples. First article test samples shall consist of 100 radiosondes representative of the production equipment. Samples shall be identified with the manufacturer's part number. Samples remaining, after the completion of all first article testing, shall be returned to the procuring activity engineering agency. Retest samples shall not be selected from the first article test samples.

4.4.2 Compatibility testing. As soon as possible after fabrication of the first article test samples has begun (at least 30 days prior to first article testing in accordance with 4.4.3), the contractor shall submit 10 radiosondes to an Air Force facility designated by the procuring activity (see 6.2) for check of interface compatibility with the AN/AMQ-25 and AN/AMQ-29 systems. An operational ground check of the radiosondes shall be included. Contractor personnel shall be permitted to witness the tests. After completion of compatibility testing the radiosondes will be returned to the contractor for any necessary modification and further testing.

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*4.4.3 Ground testing. Forty radiosondes shall be subjected to all the tests of 4.6 with the exception of the accuracy and flight tests of 4.6.12 and 4.6.13. The contractor shall prepare a test plan and test report as specified by the procuring activity.

*4.4.4 Accuracy testing. Upon approval of the test report of 4.4.3 by the procuring activity, 30 radiosondes identical to those subjected to ground testing shall be shipped to the Air Force facility designated by the procuring activity (see 6.2) for accuracy testing in accordance with 4.6.12. The procuring activity will advise the contractor of acceptance or of discrepancies found during accuracy testing. The contractor may attend and witness all testing.

*4.4.5 Flight testing. Upon successful completion of the accuracy tests, 20 sondes, identical to those used in accuracy tests, will be shipped to the Air Force facility designated by the procuring activity for flight testing in accordance with 4.6.13. The contractor may attend and witness all testing.

*4.4.6 Retest provisions. In the event the first article samples fail to meet the provisions of either 4.4.3, 4.4.4, or 4.4.5, the contractor shall take appropriate corrective action and submit additional samples for one retest. All modifications required to successfully pass the first article tests shall be incorporated in production items.

*4.5 Quality conformance inspection and testing. Quality conformance inspections will consist of two parts as follows:

*4.5.1 Production acceptance. Production acceptance shall consist of three parts:

a.	Individual tests	(4.5.1.1)
b.	Sampling tests	(4.5.1.2)
c.	Environmental tests	(4.5.1.3)

Complete records of all quality conformance testing will be maintained by the contractor and will be available for inspection by the Government upon request. The contractor shall prepare a production acceptance test plan as specified by the procuring activity. Individual tests shall be periodically monitored by the Government quality conformance representative and he shall witness and accept sampling and environmental testing.

*4.5.1.1 Individual tests. Each radiosonde shall be subjected to tests verifying the performance of 4.3.2. The test of 4.6.1 shall also be performed.

*4.5.1.2 Sampling tests. The sampling basis will be in accordance with MIL-STD-105, inspection level II, single sampling, at an AQL of 2.5 percent defective. The examination of product test as specified in 4.6.1 shall be performed.

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*4.5.1.3 Environmental tests. Environmental testing shall be performed by the contractor on not less than 10 radiosondes selected at random from each months production. The following tests shall be performed as specified in 4.6:

a. Humidity (4.6.6)
b. Rain (4.6.9)

4.5.2 Accuracy verification. The procuring activity's quality conformance representative shall select one radiosonde from every 500 delivered to be used for Government production quality conformance testing at the Government's option. These sondes shall be stored at a location specified by the procuring activity until a total of 20 sondes have been collected. No sonde will be stored in excess of the requirements of 3.5.4. Each group of 20 shall be subjected by the Government to the accuracy tests of 4.6.12. The pass/fail will be as specified in 4.6.12.5. If the radiosondes fail to meet these requirements, production and deliveries may be suspended at the Government's option until the contractor demonstrates to the satisfaction of the Government that the deficiencies which caused the failures have been corrected.

4.6 Test methods.

4.6.1 Examination of product. The radiosonde shall be examined to determine compliance with the requirements specified herein with respect to material, workmanship, marking, and dimensions.

4.6.1.1 Wiring. The wiring shall be inspected to determine that the stripping of the insulation is neat, and that not more than 10 percent of the strands, or one strand, whichever is greater, have been broken. The insulation shall be intact and show no evidence of burns or tears. The wiring shall be checked for proper coding, proper crimping of the wire at the terminals, flexibility of pigtails and uniform layout of the wiring.

4.6.1.2 Soldering. All soldered connections shall be inspected. Acceptance shall be based on all connections being neat, smooth, and intact and the soldering feathering out to a thin edge. Cold or rosin joints shall be rejected. Thermoplastic material shall be examined for evidence of overheating and distortion due to soldering operations.

4.6.1.3 Rigidity of assemblies. All rivets, screws, eyelets, and other fastening devices shall be inspected to determine that they are firmly seated and hold the parts firmly.

4.6.1.4 Plastic parts. Molded plastic parts shall be inspected to see that they are clean of flash, free from cracks, and possess a clean uniform surface. Laminated plastic materials shall be checked for treatment with moisture-resistant material.

4.6.1.5 Moisture proofing. Moisture proofing provisions shall be inspected to see that they are complete and undamaged.

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4.6.2 Low temperature operation. The radiosonde shall be placed in a test chamber at a temperature of 0°C ± 2°C for a period of 4 hours. At the end of this period the temperature shall be reduced to -65°C ± 2°C. The radiosonde shall be operated from internal batteries for a period of 5 minutes, during which the radiosonde shall be tested for proper operation in accordance with 4.3.2. Within 2 minutes of the completion of this test, the radiosonde shall be placed in an environment at +50°C ± 5°C with relative humidity above 90 percent. The radiosonde shall be operated for an additional 5 minutes and tested for proper operation in accordance with 4.3.2.

4.6.3 Low temperature storage. The radiosonde shall be stored for 24 hours at -60°C ± 5°C. The temperature shall be raised to 25°C ± 5°C within one hour. When the radiosonde temperature has stabilized it shall be operated from internal batteries for 10 minutes and tested for proper operation in accordance with 4.3.2.

4.6.4 High temperature operation. The radiosonde shall be placed in a test chamber at 50°C ± 2°C for 3 hours with no power applied. At the end of this period and while at this temperature, the radiosonde shall be operated from internal batteries for 10 minutes and tested for proper operation in accordance with 4.3.2.

4.6.5 High temperature storage. The radiosonde (nonoperating) shall be stored at 72°C ± 2°C for 24 hours. The temperature shall then be lowered to 25° C ± 5°C within 1 hour. When radiosonde temperature has stabilized, it shall be operated from internal batteries for 10 minutes and tested for proper operation in accordance with 4.3.2.

4.6.6 Humidity. The radiosonde shall be positioned in a humidity chamber where the relative humidity shall be maintained at 95 percent for a period of 44 hours. The chamber temperature shall be varied from room temperature to 50°C ± 2°C and back to room temperature in two 22 hour cycles. The radiosonde shall then be operated from internal batteries for 2 minutes. Within 10 minutes of the conclusion of temperature cycling the radiosonde shall be removed from the chamber, any excess moisture removed by turning the radiosonde upside down (external surfaces may be wiped), and the radiosonde tested for proper operation in accordance with 4.3.2.

4.6.7 Vibration. The radiosonde, completely assembled (including sensor elements but not operating) shall be mounted on a vibration stand and subjected to vibration at an amplitude of 0.02 ± 0.002 inch (Total excursion of 0.04 inch). The frequency of vibration shall be cycled from 10 Hz to 55 Hz and back to 10 Hz at a rate of one sweep per minute. The radiosonde shall be vibrated for 1 hour in each of three mutually perpendicular planes. At the end of this period the radiosonde shall be operated from internal batteries for 5 minutes and tested for proper operation in accordance with 4.3.2. Only a total of four radiosondes shall be subjected to this test.

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*4.6.8 Explosion. Explosion testing shall be in accordance with MIL-STD-810, method 511, procedure I. A total of four radiosondes shall be subjected to this test.

4.6.9 Rain. The radiosonde shall be subjected to a combination of 2 inches per hour simulated rainfall and 100 feet per second wind into the sensor end and an additional 2 inches per hour simulated rainfall which is 30 degrees off the radiosonde centerline, near the parachute end. This rain test shall be applied for 2 minutes with the radiosonde operating in its normal drop attitude. The radiosonde shall continue to operate 8 minutes after the wind and water have been turned off, during which it shall be tested for proper operation in accordance with 4.3.2.

*4.6.10 Acceleration. The radiosonde shall be subjected to a nonoperative acceleration test at a maximum acceleration of 5 g's along the centerline of the cylinder. The radiosonde shall be placed in the centrifuge in such a position that 90 percent of the unit under test shall be subjected to the specified G level. The test time shall be 1 minute. After acceleration the radiosonde shall be operated from internal power for 5 minutes and tested for proper operation in accordance with 4.3.2. A total of ten radiosondes shall be subjected to this test.

*4.6.11 Shock. The radiosonde, less parachute (if equipped) shall be prepared with both sensors installed and the antenna extended. The radiosonde, with internal power applied, shall be suspended by the parachute mounting blocks and subjected to a shock of 75 g's along the centerline of the cylinder. The acceleration shall be measured with a suitable instrument. The radiosonde and sensor elements shall be visually inspected for damage or loosening. The radiosonde shall then be operated from internal power for 5 minutes and tested for proper operation in accordance with 4.3.2. A total of ten radiosondes shall be subjected to this test.

*4.6.12 Accuracy test. Thirty sondes shall be tested for accuracy in an environmental wind tunnel at the Arnold Engineering Development Center (AEDC) or another such facility as approved by the procuring agency.

*4.6.12.1 Tunnel installation. The radiosondes will be installed in the tunnel with the long axis parallel to the direction of the air flow and the sensor ends facing upstream. A total of four sondes will be mounted in the test section for each test run. The parachute housing and antenna will be removed to prevent RF transmission.

*4.6.12.2 Tunnel instrumentation.

*4.6.12.2.1 Arrangement. All instrumentation and wiring will be arranged such that it does not interfere with the air flow around the sonde sensors. Sonde power will be provided by precision power supplies.

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*4.6.12.2.2 Connection. The connection of instrumentation to the sondes will meet the following criteria:

a. The sonde data recorded will be in the form of AF oscillator output audio frequency.

b. The sonde transmitter will be made inoperative in a manner that will not interfere with the audio circuit performance.

c. The sonde audio circuit will not be loaded due to low impedance at the instrumentation input.

*4.6.12.2.3 Tunnel sensors. The tunnel temperature, pressure and humidity sensors will be placed as close to the sonde sensors as possible consistent with the requirements of 4.6.12.2.1. The total (3 sigma) uncertainty in the tunnel instrumentation, including the combined sensor, cross-sectional flow nonuniformity, data acquisition and data reduction errors, will not exceed the following limits:

a. Temperature ± 0.25°C
b. Relative humidity ± 5% RH
c. Static pressure ± 1 mb (0.015 psi)

*4.6.12.3 Baseline procedures. All baselining for the accuracy tests will be done in the closed wind tunnel to prevent the possibility of introducing errors in the radiosonde data due to baseline inaccuracies. The sonde audio frequency output will be used to determine the sensor resistance. The tunnel sensor output will then be used as the standard for the calculation of the baseline correction factor.

*4.6.12.3.1 Pressure baseline. The pressure sensor will be baselined with no airflow and at ambient pressure.

*4.6.12.3.2 Temperature baseline. The temperature sensor will be baselined with air flowing at 75 ± 15 feet persecond and at ambient tunnel temperature.

*4.6.12.3.3 Relative humidity baseline. The baseline resistance value supplied with the individual humidity sensors will be used in the calculation of the relative humidity baseline correction factor.

*4.6.12.4 Test procedures. Each sonde will be subjected to the following series of tests. All tests will be conducted with the sondes commutating normally. All tests except the pressure test will be with air flowing at 75 ± 15 feet per second.

*4.6.12.4.1 Pressure test. The pressure test will be conducted with no airflow in the tunnel. The pressure will be lowered from ambient to 225 ± 25 mb in 2 ± 0.5 minutes, then raised to ambient in 8 ± 1 minutes.

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*4.6.12.4.2 Relative humidity test. Two relative humidity tests will be conducted.

a. The first test will be at 500 ± 50 mb pressure and -10 ± 5°C. The relative humidity will be raised in steps of approximately 10% RH magnitude from ambient to 90 ± 5% RH, then returned to ambient in the same manner. At least 15 seconds will be allowed to elapse at each level between steps. The total sequence must take less than 9 minutes.

b. The second test will be at 850 ± 50 mb pressure and ambient temperature. The relative humidity will be raised from 30 ± 10% RH to 95 ± 5% RH in steps of approximately 10% RH magnitude, then returned to 30 ± 10% RH in the same manner. At least 15 seconds will be allowed to elapse at each level between steps. The total sequence must take less than 9 minutes.

*4.6.12.4.3 Temperature test. Two temperature tests will be conducted, both at 700 mb pressure.

a. The first test will be used to determine sonde accuracy at low temperature rates of change and to determine the effect of the temperature shock encountered at launch on radiosonde accuracy. The temperature will be changed from ambient to -20 ± 5°C in 1.0 ± 0.5 minutes and returned to ambient at a rate not to exceed 10°C per minute. The total sequence must take less than 8 minutes.

b. The second test will be used to determine temperature accuracy at higher temperature rates of change. The temperature will be varied between +10 ±5°C and -10 ± 5°C at various rates of change between 10°C per minute and 35°C per minute. The total sequence must take less than 12 minutes.

*4.6.12.5 Failure criteria. Any sonde for which more than 10% of the data points do not meet the following criteria will be considered a failure. Failure of more than 10% of the sondes tested shall constitute failure of the accuracy testing. All accuracy requirements include the allowance for tunnel sensor uncertainty.

a. Pressure ± 3.5 mb (for increasing pressures only; no requirement for decreasing pressure).

b. Relative humidity: ± 7% RH at all humidities between 60 and 100% RH and ± 10% RH at humidities below 60% RH at all temperatures between -40°C and +50°C.

c. Temperature: ± 1.0°C at all temperature time rates of change less than the atmospheric equivalent of 4.0°C per 1000 ft; ± 1.5°C at rates between 4.0°C and 8.0°C per 1000 ft; ± 2.0 °C at rates between 8.0°C and 12.0°C per 1000 ft.

*4.6.13 Flight testing. Upon successful completion of the accuracy testing, 20 sondes will be subjected to the following flight test. The sole purpose of this test will be to check the reliability of the sonde in the operational environment.

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*4.6.13.1 Test mission. The sondes will be baselined using the normal operational techniques and dropped from a WC-135 aircraft on a normal operational mission. All 20 sondes need not be dropped on the same mission. The sondes will be launched at normal aircraft cruise speed at an altitude of 300 mb.

*4.6.13.2 Failure criteria. At least 90% of the sondes must successfully pass the operational baseline check and transmit usable data continuously from launch to impact at the surface. In any case where data from a sonde is lost due to aircraft equipment malfunction, the sonde will not be included in the total used for computation of the pass/fail percentage. The Air Force test director will be the sole judge of whether the aircraft systems were at fault. At least 15 sondes must be included in the total for calculation of the pass/fail percentage for the test to be valid.

*4.6.14 Antenna radiation pattern. The radiation pattern of the radiosonde antenna shall be obtained in the following manner: The radiosonde shall be fully assembled and powered by its internal battery. The radiosonde shall be mounted at least 15 feet above the ground on a level terrain area, with no obstruction within 100 feet of the radiosonde site. The radiosonde shall be in the post-launch configuration. A receiver and antenna shall receive the radiosonde signal and the receiver signal intensity versus azimuth angle shall be plotted on polar chart paper. Two orthogonal E-Plane patterns and one H-Plane pattern shall be obtained. The patterns shall meet the requirements of 3.7.4.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Levels A and B. Unless otherwise specified, cleaning, drying, preservation and packaging shall be in accordance with MIL-P-116. The selection of applicable methods shall be in accordance with MIL-E-17555.

5.1.2 Level C. Unless otherwise specified, radiosondes shall be preserved and packaged with sufficient protection to assure the product is acceptable at destination.

5.2 Packing.

5.2.1 Level A. Exterior containers shall be overseas type in accordance with the applicable specification. Insofar as practicable, exterior containers shall be of uniform shape and size and shall contain identical quantities.

5.2.2 Level B. Exterior containers should be the same as referenced for Level A; however, a domestic type container in accordance with the applicable specification is authorized for this level. Insofar as practicable, exterior containers shall be of uniform size and shape and shall contain identical quantities.

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5.2.3 Level C. Packing shall afford protection against damage during domestic shipment from source to the first domestic receiving activity for immediate use. This level in general will conform to the applicable carriers rules and regulations and may be the supplier's commercial practice, provided the latter meets the requirements of this level.

5.3 Reinspection marking. Reinspection marking shall be as specified by the procuring activity.

*5.4 Humidity sensor. The humidity sensor shall be packed in a hermetically sealed can. The can will be marked in accordance with MIL-STD-130 and shall include the date of manufacture and the data specified in 3.7.1.3.

6. NOTES

6.1 Intended use. Radiosonde AN/AMT-13 ( ) is intended for use in the collection of atmospheric meteorological data. It is intended for launching aft from a high-altitude high-performance aircraft.

*6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Point of delivery for first article compatibility samples (see 4.4.2), accuracy test samples (see 4.4.4), first article flight test samples (see 4.4.5) and accuracy verification samples (see 4.5.2).

c. Baselining procedure to be followed prior to flight test (see 4.6.13).

d. Level of packaging and packing required (see section 5).

e. Provisions for Government-loaned property (see 3.12).

f. Requirements for test plans and test reports.

g. Type of battery to be supplied as GFE.

NOTE: The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notiations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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